Group Project Proposal

DARA: Debate, Adjudicate, Recommend, Assess

Group ID: 04

DSA4213: Natural Language Processing for Data Science
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Administrative Information

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Project direction	The multi-agent system enables Singapore to rehearse the negotiation in advance, anticipating the positions, constraints, and objections it is likely to encounter.
Repository	

Project Summary

We build a debate-based multi-agent system that supports Singapore's crisis mitigation planning by simulating its negotiations with regional counterparts. LLM agents representing different countries argue competing mitigation strategies over three rounds; an independent judge agent scores each round against a fixed rubric; and a recommender agent converts the judge's assessments into conditional, priority-tagged negotiation options for Singapore. We evaluate against a single-model baseline and a no-judge ablation, measuring whether each country-agent stays consistent with its assigned interests, whether the judge scores reproducibly, and whether human evaluators find that the final recommendation fairly incorporates the strongest points from both sides.

1. Problem, Research Question, and Motivation

Research question. Can a debate-based multi-agent system, in which LLM agents representing Singapore and its counterpart countries negotiate over multiple rounds before an independent judge and recommender synthesise the outcome, help Singapore rehearse crisis negotiations (anticipating counterparts' positions, constraints, and likely objections) and produce mitigation recommendations that, compared with a single-model response, better advance Singapore's priorities while remaining acceptable to other stakeholders?

Motivation. Real-world crisis response, whether to a natural disaster, a cybersecurity breach, or a public health emergency, rarely involves a single decision-maker. It typically involves multiple countries or stakeholders with different resources, priorities, and constraints (a wealthier nation may prioritise speed and cost, a smaller or more vulnerable nation may prioritise equity or direct aid, and so on), and the eventual response is a negotiated compromise rather than one party's ideal

plan. Static, single-domain decision-support tools do not capture this negotiation: they produce one recommendation without ever weighing it against opposing positions. This matters practically for governments that must reconcile competing national interests under time pressure. It matters scientifically because it tests whether LLM agents can sustain a structured, adversarial argument over multiple rounds, and whether an independent LLM judge can meaningfully evaluate competing proposals rather than defaulting to superficial or biased judgments.

Why agent-based. A single model asked to "give the best crisis response" collapses every country's perspective into one voice, so the underlying trade-offs and disagreements are hidden rather than resolved. A debate structure makes those trade-offs explicit and inspectable: each country-agent is responsible only for building the strongest case for its own position, the judge agent is responsible only for evaluating the arguments on a defined rubric, and the recommender agent is responsible only for synthesising a final solution from the judge's output. Separating these roles across rounds also lets each agent revise its position in response to the others, so the final recommendation reflects deliberation rather than a single pass of reasoning.

Project scope. The project simulates a bilateral crisis negotiation from the perspective of Singapore, the country of interest. In each scenario, one country-agent represents Singapore and another represents a single counterpart nation, drawn primarily from Southeast Asia. Each agent is briefed with its country's publicly stated interests, resources, and constraints. The two agents debate mitigation strategies for the crisis over up to three rounds, with each round building on the arguments raised in the previous one.

Intended users. Policymakers and government officials in Singapore preparing negotiation positions ahead of or during a regional crisis, and researchers studying multi-agent negotiation who want a tool to stress-test proposed mitigation strategies against opposing viewpoints.

Definition of success. The project succeeds if:

(a) each country-agent produces arguments consistent with its assigned interests and constraints across all three rounds, backed by relevant sources (i.e. it does not drift into a generic, uncommitted position);

(b) the judge's evaluations are consistent, independent, and justifiable against a predefined rubric rather than arbitrary;

(c) a recommender agent, which has no equivalent in the prior work reviewed in Section 2, converts the judge's assessments into a set of conditional, priority-tagged mitigation options for Singapore, rather than a single generic verdict; and

(d) on past regional crises, the system produces accurate and robust recommendations that align with the policies actually adopted.

2. Related Work and Proposed Contribution

Multi-agent debate for LLM reasoning. The core mechanism of our system, having multiple LLM agents argue competing positions before a separate evaluator, builds directly on multi-agent debate (MAD) research. Du et al. [1] first showed that having multiple LLM instances generate,

critique, and revise each other's answers over several rounds improves factual accuracy and reasoning on tasks such as math and strategic decision-making, and Liang et al.'s MAD framework [2] extended this with an explicit judge model supervising the debate to counteract "degeneration of thought," where debating agents converge prematurely on the same answer instead of genuinely disagreeing. The idea of a judge overseeing debating agents traces back further to Irving et al.'s AI Safety via Debate [3], which argues that adversarial arguments in front of a judge can make correct answers easier to identify than to generate outright. More recent work is more cautious: Smit et al. [4] and related studies find that MAD's benefits are highly sensitive to configuration (number of rounds, agent diversity) and can be undermined by a poorly calibrated judge, which is directly relevant to our three-round design and motivates treating the judge's calibration as a first-class design concern rather than an afterthought.

LLM-as-judge. Our independent judge agent draws on the LLM-as-judge paradigm established by Zheng et al. [5], who showed that a strong LLM can approximate human preference judgments when scoring the outputs of other models on a defined rubric. This approach has since been extended specifically to debate settings [2], where a judge must score argument quality and consistency across multiple rounds rather than a single response.

LLM agents in negotiation and diplomacy. Because our scenario is explicitly a bilateral negotiation rather than abstract reasoning, we also draw on LLM negotiation-agent research. Meta AI's Cicero [6] demonstrated that a language model combined with strategic planning could negotiate at human level in the game of Diplomacy, and follow-up work such as Richelieu [7] showed this could be achieved with a self-evolving LLM agent alone, without Cicero's separately trained reinforcement-learning planner. Closer to our interest-driven political framing, Moghimifar et al. [8] modelled multi-party political coalition negotiations using LLM-driven dialogue, and the Political-LLM survey [9] documents related work simulating legislative debate and cross-national alliance-building. These systems establish that LLM agents can sustain realistic, interest-driven negotiation dialogue, but none of them combine that with a separate judge scoring argument quality or a downstream agent that turns the outcome into actionable, priority-conditioned guidance for one side.

Gap and contribution. Existing multi-agent debate systems [1, 2] are evaluated on abstract reasoning or QA benchmarks, not on realistic, interest-driven negotiation. Existing LLM negotiation and diplomacy agents [6, 7, 8] simulate realistic back-and-forth bargaining, but do not incorporate an independent judge scoring the exchange, nor do they produce an explicit, decision-ready output for one party. Our contribution is to combine these two lines of work into a single pipeline purpose-built for crisis mitigation:

1. country-agents grounded in stated national interests, resources, and constraints (in the spirit of Cicero/Richelieu-style negotiation agents) debate a crisis scenario over three rounds;
2. an independent LLM judge, following the LLM-as-judge and MAD-supervisor paradigms, scores each round against a predefined rubric covering argument feasibility and trade-offs rather than simply picking a "winner";
3. a recommender agent, which has no equivalent in the prior work reviewed above, converts the judge's assessments into a set of conditional, priority-tagged mitigation options for Singapore, rather than a single generic verdict; and
4. unlike fully autonomous debate or negotiation pipelines, a human reviewer confirms the issue

set, options and weights before any agent runs (Figure 1), and can inspect and, where necessary, intervene at the judge's assessment stage or before the recommender's final output is issued, so the system functions as a decision-support aid rather than an unsupervised decision-maker.

This closes the gap between debate-for-reasoning systems and negotiation-simulation systems, neither of which currently produces actionable, party-specific decision support.

3. Agent and Task Setup

Task and environment. We study a multi-agent system for crisis mitigation in a bilateral setting. Each scenario describes a crisis involving Singapore as the focus country and one counterpart country. Examples may include natural disasters, cybersecurity incidents, or public health emergencies where the countries need to coordinate a response despite having different priorities and constraints. Every participating country is given a country brief specifying its interests, available resources, constraints, and red lines. These briefs define the perspective that each country-agent is expected to maintain throughout the debate. All country-agents use the same underlying prompt structure; only their country briefs differ.

Agent roles. The system contains three agent roles:

(a) **Country-agents.** Each country-agent represents one country involved in the crisis. Its task is to propose and defend mitigation strategies that are consistent with its assigned brief. An agent's argument includes its position, supporting claims and evidence, possible compromises, and red lines. The agents are expected to advocate for their own country's interests rather than immediately converging on a generic compromise.

(b) **Judge.** The judge is an independent LLM that does not receive a country brief and therefore has no assigned national position. After each debate round, it evaluates the agents using a predefined rubric covering feasibility, specificity, acknowledgement of trade-offs, consistency with the agent's own previously stated position, and responsiveness to opposing arguments. The judge provides scores with written justifications and identifies important trade-offs that have not yet been addressed.

(c) **Recommender.** The recommender produces the final decision-support output for Singapore and does not participate in the debate. In the full system, it receives the crisis scenario and the judge's assessments across all three rounds, but not the raw debate transcript. It uses these assessments to recommend a final settlement package for Singapore while explaining the relevant trade-offs.

Interaction protocol. The country-agents debate for three rounds. In Round 1, each agent receives the common crisis scenario and its own country brief and independently presents its initial position. In Rounds 2 and 3, the agents additionally receive the opposing arguments and judge feedback from the previous round. This allows them to rebut opposing claims, respond to weaknesses identified by the judge, and revise their proposals while remaining consistent with their assigned interests. The judge evaluates the arguments after every round using the same rubric.

After Round 3, the debate ends and the three judge assessments are passed to the recommender. The recommender then produces the final set of mitigation and negotiation options for Singapore.

Inputs, evidence, and outputs. Each run consumes one structured crisis scenario, stored as JSON, with six parts (Table 1).

Table 1: Components of the input.

Part	What it holds
Parties	The focus country (Singapore) and one counterpart.
Issues and options	The matters under negotiation and the discrete settlement levels available for each. This is the menu: nothing outside it can be agreed.
Briefs	Per country: interests, resources, constraints, and red lines, with each element recorded alongside its source.
Conditions	Current circumstances that limit what a party can *offer*, such as a committed budget, an election, or a concurrent disaster. Dated and pinned for the run, so that compared runs can be verified to have read the same world.
Valuations	Per country: a weight for each issue and a value for each option. Private: visible only to that country-agent, and withheld from the opposing agent and the judge.
Evidence cutoff	The date beyond which no material may be used.

For testing, we use past crisis cases. We author the six parts ourselves from the public record, admitting nothing published after a cutoff that precedes the outcome, so that the settlement the parties actually reached can serve as a reference point. In real use, the user supplies the same six parts for a crisis that is still open, either by entering them directly or by having them extracted from documents they provide. In that setting, no such reference exists.

Per round, each country-agent returns a structured argument: its position, claims, proposed compromises, red lines, and a settlement package naming one option per issue. From Round 2 it also receives the opposing arguments, the judge's feedback on its own previous turn, and any trade-offs the judge recorded as unaddressed. The judge returns a structured assessment after every round, and after Round 3 the recommender converts those assessments into advice for Singapore; in the full-system condition it does not receive the raw transcript. That advice is a headline, a set of *conditional options*, and one opening package. Each option names the condition it applies under, the action to take, the concession that action requires, and its principal risk, and carries a *priority tag* drawn from a fixed list: cost, speed, equity, casualty minimisation, or sovereignty. The tags come from a closed set so that the recommender cannot invent a category the evaluation would then silently drop, and the opening package is what is scored against the benchmark.

Settlement packages are checked against the scenario's legal options; invalid packages are recorded as errors rather than silently corrected. The scenario, arguments, assessments, and recommendation are all written to a run log.

Study conditions. To test whether the debate structure provides benefits over a direct LLM response, we evaluate several system conditions on the same set of crisis scenarios. The main comparison is between a **single-model baseline**, which directly generates a crisis response for Singapore, and the **full multi-agent system**, which uses up to three rounds of debate followed by judging and recommendation. We additionally consider a debate condition without the judge, different numbers of debate rounds, and variations in the country briefs. These conditions allow us to separate the contributions of the debate, the judge, and the diversity of country interests, rather than treating the full pipeline as a single black box. The exact comparisons and ablations are described in Section 4.1.

Stopping conditions and constraints. Each debate runs for a maximum of three rounds, after which the recommender produces the final output. We cap the debate at three rounds for three reasons. First, three rounds is the minimum that allows a full exchange: an opening position, a response to the counterpart's arguments, and a revised position that incorporates the judge's feedback. Second, prior work on multi-agent debate suggests that improvements tend to diminish after a small number of rounds, while longer debates risk repetition and premature convergence towards generic, overly agreeable positions. Third, each additional round increases token usage, latency, and context length, which matters for a tool intended to support negotiation preparation under time pressure. This cap is treated as a hypothesis rather than a fixed assumption: the round-count ablation compares one, two, and three rounds, and fewer rounds are adopted if they provide comparable deliberation and recommendation quality.

4. Proposed System or Study Design

Section 3 lists the six parts of a crisis scenario and the three agent roles. This section describes where those inputs come from in the study, and the order in which the pipeline uses them.

Testing versus real use. For **testing** we use a closed historical case: we author every input from the public record, admit nothing published after a declared evidence cutoff, and, because the episode is over, also hold what the parties actually settled on, which is what makes a backtest possible. For **real use** the crisis is still open: the user supplies the inputs, and no ground truth exists to score against. The study set is two cases, both from Singapore's perspective: transboundary haze with Indonesia (2015–16) and COVID-19 border and supply measures with Malaysia (2020). Each has four issues at three settlement levels, so $3^4 = 81$ settlements are possible and nothing outside that set can be agreed.

Table 2: Where each input comes from. In testing we author both valuation tables, so the payoff structure is exact; in real use the counterpart's valuations are the user's estimate.

Input	**Testing (we author)**	**Real use (user supplies)**
Parties	Singapore and one counterpart	User names both
Issues & options	Authored from the record (*carries hindsight*)	Typed, or extracted from pasted documents
Own valuations	Authored	User gives a weight per issue and a value per option
Counterpart valuations	Authored	**User estimates them**; the weakest input in the system
Briefs	From public statements, under the cutoff	Extracted from pasted documents, or typed
Conditions	Frozen in the case file, all dated $\leq$ cutoff	Fetched once per session, then pinned
Evidence cutoff	Set per case, before the outcome	User sets; defaults to today
Ground truth	The settlement actually reached (the status quo where none was)	**None exists**

The pipeline. Figure 1 summarises the six stages of a run, from assembling the problem through to stress-testing the conclusions.

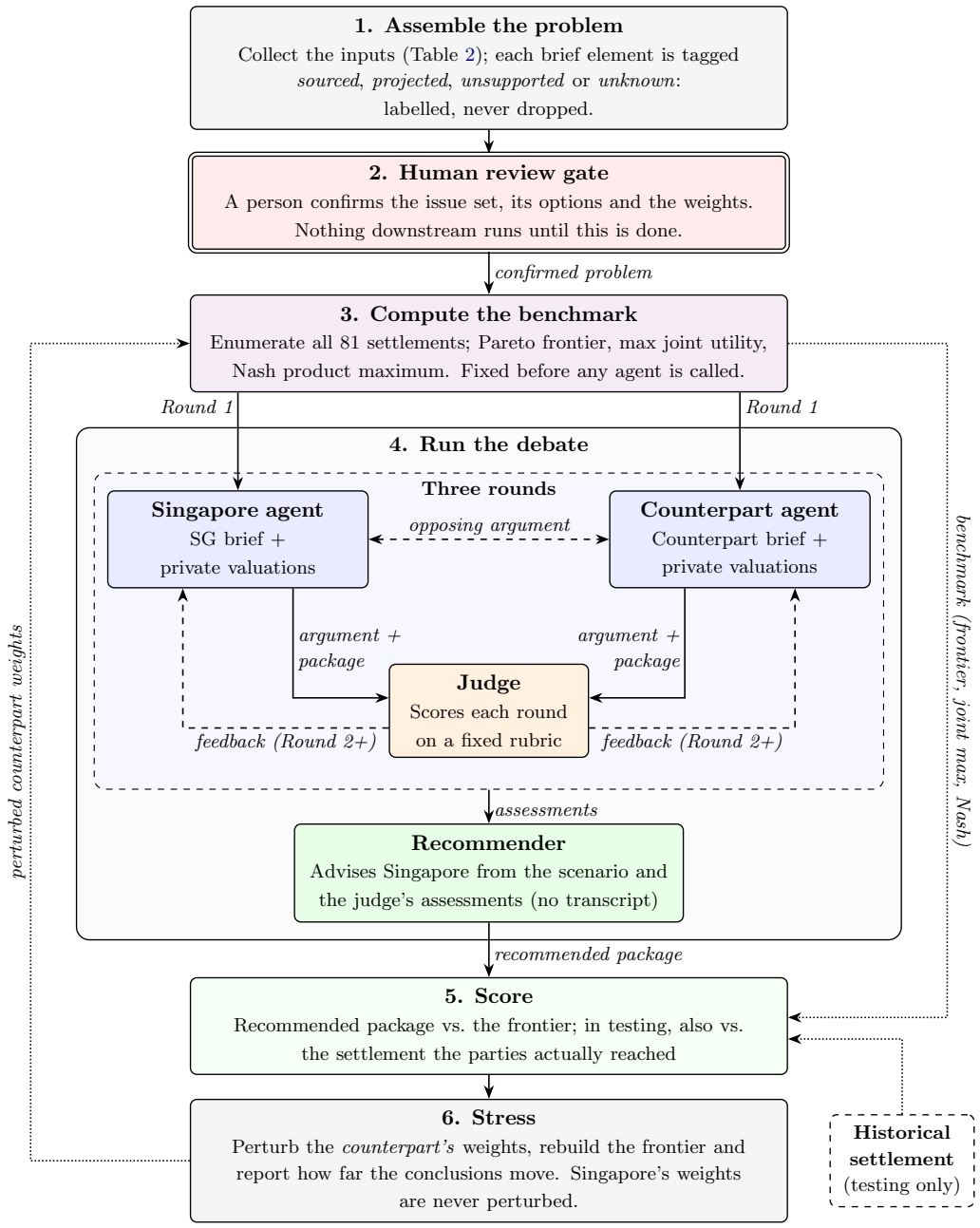


Figure 1: Overall pipeline. The problem is assembled with provenance labels and confirmed by a human before anything else runs. The benchmark is computed by enumerating all settlements before any agent is called, so it cannot move to fit the result. The agents then debate over three rounds, and the recommender advises Singapore from the scenario and the judge's assessments, without the transcript. The recommended package is scored against the frontier (and, in testing, the historical settlement), and the conclusions are stress-tested by perturbing the counterpart's weights.

Why the benchmark is a joint one. In the haze case, 14 of the 81 settlements are Pareto-optimal, and the joint-maximising settlement reaches 1.358 against 1.048 for splitting every issue down the middle, so 0.310 of joint utility is what skilled negotiation is worth here. Pareto efficiency is a *waste* criterion rather than a fairness one, which is why a Singapore-facing tool uses it: of the 67 dominated settlements, all 67 admit an alternative strictly better *for Singapore*, and Singapore's own best available deal (1.000, against Indonesia's 0.062) is itself on the frontier. We nonetheless score the research question on joint value recovered, because scoring on Singapore's utility alone would reward *demand everything*, which is reachable in one call with no debate and no discovery of what the counterpart values, and so could not distinguish the conditions of Section 4.1.

What the group owns. Our contribution lies not in the model calls themselves but in three layers. (i) The **payoff layer**: requiring every argument to commit to a concrete package, which converts "this recommendation is better" from a reading into an arithmetic claim and is what makes a negative result possible at all. (ii) The **information architecture**: agents blind to each other's valuations, a judge blind to all valuations, and a recommender blind to the transcript, enforced by the interfaces between stages rather than requested in prompts, so a barrier cannot be lost to a prompt edit. (iii) The **contamination and provenance layer**: both cases are historical, so an evidence cutoff, a leak scan and per-element provenance are what keep a backtest from silently measuring recall.

4.1 Comparisons and Ablations

All three conditions receive the same scenario, the same briefs and the same decoding settings, and differ only in how the debate is structured, so a difference between them is a difference in structure rather than in the problem they were given.

Table 3: The three experimental conditions. Call counts are reported with every comparison, because the budget gap is a declared limitation rather than a control.

Condition	Calls	Structure
B0	2	Single-model response; no debate, no judge
B1	8	Three debate rounds; recommender reads the raw transcript
B2	11	Three rounds, judge each round, transcript-blind recommender

Primary comparison. The primary comparison is B2 against B0. The budget gap is a declared limitation, not a control: B2 spends 11 calls against B0's 2, so a B2 win is confounded with having spent more, and the honest reading is "structure *or* budget". B1 is the internal discriminator, with 8 calls and a debate but no judge, so B2 > B1 isolates the judge (together with the transcript-blind recommender) at near-equal cost.

Non-model baselines. *Split-the-difference* (every issue at its middle option, joint 1.048) is what a process that discovers nothing should produce; a condition that cannot beat it has found no trade, however its transcript reads. *The historical settlement* is sharper. The 2015–16 haze dispute ended in no agreement, so we score it as the status quo on each issue; scored against our frontier that outcome is worth 0.268 to Singapore and 0.660 to Indonesia: joint 0.928, not Pareto-optimal,

0.430 below the best jointly available, and *worse for Singapore than halving every issue*. The real parties left value on the table, which is both the motivation for the tool and a floor no condition should fail to clear.

Ablations beyond those in Section 5. Section 5.1 covers the round-count ablation and Section 5.3 the judge robustness tests. Four further ablations isolate design decisions specific to this pipeline, holding seeds and scenario fixed.

- **Brief diversity.** Give every country-agent an identical brief and identical valuations. If the debate still looks adversarial, the disagreement is sampling noise rather than assigned interests, which would undercut success criterion (a).
- **Style on/off.** Per-country register and sentence shape, supplied to each country-agent as a short *style file*, with a leak check asserting that no issue name or option value appears in any style file: style may carry *how* a party speaks, never *what* it wants.
- **Conditions snapshot.** Situational constraints are pinned before the benchmark is computed and their identifier recorded, so two compared runs can be verified to have read the same world; a stale-snapshot arm reports how far the advice moves when that world model is a day older.
- **Information-transmission probe.** Compare integrative gain under private valuations against an upper bound in which both agents see both valuation tables. The gap is the joint value the dialogue failed to transmit, which isolates argumentation from arithmetic.

Contamination controls. Because both cases are historical, four probes of increasing cost ask whether the model is recalling rather than reasoning: a purely local *leak scan* (P0) for post-cutoff markers absent from every supplied input but present in the output, then *free recall* (P1), *cued prediction* (P2), and *discrimination* (P3) between the real outcome and a plausible fabrication, where scoring above chance is recall. P0 is the sharpest despite being cheapest, because it tests the system as built rather than the model in the abstract. A subtler leak runs through the *issue set*, authored by people who knew how the negotiations ended; until issue sets are derived wholly from pre-cutoff documents, absolute backtest numbers are reported as contaminated. Relative comparison between B0, B1 and B2 remains valid, since all three face the same menu, and that comparison, not absolute agreement with history, is what we report.

5. Evaluation

5.1 Round-Count Ablation and Debate Trajectory Analysis

Before evaluating the final system, we determine the optimal number of debate rounds by comparing 1, 2, and 3 rounds across both crisis scenarios. Our hypothesis is that 3 rounds may provide sufficient deliberation, but additional rounds should only be retained if they yield meaningful improvement. We select the smallest number of rounds that achieves meaningful deliberation and recommendation quality, since fewer rounds reduce latency and computational cost.

We evaluate the metrics in Table 4.

Table 4: Evaluation metrics for the number of debate rounds.

Metric	Definition	How it is measured
Position Change	Degree to which an agent changes its position or argument between consecutive debate rounds.	Compute sentence embeddings for each agent's response and calculate cosine similarity between consecutive rounds (R1–R2 and R2–R3). Lower similarity indicates greater semantic change in the response.
Novel Argument Rate	Extent to which later rounds introduce substantive new arguments, evidence, constraints, or trade-offs rather than repeating earlier points.	A separate evaluator LLM identifies and classifies arguments in each round as new or previously raised. We calculate: $\frac{\text{Number of new substantive arguments}}{\text{Total arguments in current round}}$
Role Adherence	Degree to which each country-agent continues to represent its assigned national interests, priorities, and constraints.	A separate evaluator LLM scores each response using a predefined 1–5 role-adherence rubric, given the country's assigned profile and constraints. A subset of responses may be manually evaluated for validation.
Agent Convergence / False Consensus	Degree to which opposing agents converge toward similar positions despite unresolved differences in their assigned interests.	Track cross-agent semantic similarity across rounds and combine this with role-adherence and stance evaluation. Increasing agreement or similarity alongside declining role adherence indicates possible false consensus.
Marginal Recommendation Improvement	Improvement in final recommendation quality gained from adding another debate round.	Generate recommendations after 1, 2, and 3 rounds and evaluate each using the same recommendation rubric. Compare the change in scores from R1→R2 and R2→R3.
Cost and Latency	Additional computational resources required as the number of debate rounds increases.	Record input/output token usage, total tokens, estimated API cost, and end-to-end execution time for the 1-, 2-, and 3-round configurations.

Quantitative measures such as embedding-based position change, token usage, and latency will be computed automatically. Qualitative properties requiring contextual interpretation, including novel argument introduction, role adherence, substantive convergence, and recommendation quality, will be assessed using a separate evaluator LLM with fixed scoring rubrics. A subset of LLM-based evaluations will be manually reviewed to assess evaluator reliability.

5.2 Debater / Country-Agent Evaluation

Each crisis scenario is run multiple times using the same configuration. Complete debate sessions are cached before evaluation, rather than evaluating only one live session.

Table 5: Evaluation metrics for country-agent performance.

Metric	Definition
Role Consistency	Degree to which the agent's arguments remain consistent with its assigned national interests, priorities, resources, and constraints.
Crisis Relevance	Degree to which the agent's arguments directly address the specific crisis and its key issues.
Feasibility	Degree to which the agent's proposed actions are realistic given the country's stated resources and constraints.
Responsiveness	Degree to which the agent meaningfully addresses arguments, objections, or proposals raised by the opposing country-agent in the previous round.
Evidence Use	Degree to which supporting evidence is relevant and appropriately used to justify the agent's claims.

Scoring. Each metric in Table 5 is scored on a 1–5 scale by a separate evaluator LLM using a fixed rubric, and scores are averaged across rounds and repeated runs. A subset of responses will be human-evaluated using the same rubric, and these scores will be compared with the LLM evaluator's to check whether automated evaluation is reasonably reliable. The results are used to determine whether agents maintain distinct stakeholder positions rather than drifting towards generic or overly agreeable responses.

5.3 Judge-Agent Evaluation

The judge is evaluated on the metrics in Table 6 and the robustness tests in Table 7.

Table 6: Evaluation metrics for the judge.

Metric	Measurement
Score Agreement	Compare judge scores with independent reference scores using Mean Absolute Error (MAE) and Spearman correlation.
Ranking Agreement	Compare the judge's ranking of country arguments with reference rankings using pairwise agreement rate or Spearman correlation.
Fact-Checking Accuracy	Evaluate the judge's identification of unsupported claims using precision, recall, and F1-score against reference annotations.
Critique Coverage	Measure the proportion of reference strengths, weaknesses, feasibility constraints, and major trade-offs correctly identified by the judge.
Judgment Stability	Measure score variance and mean absolute score differences across repeated-judgment, order-swap, and country-label tests.

Table 7: Robustness tests for the judge.

Robustness test	Procedure	Measurement
Order-Swap Test	Reverse the presentation order of country arguments while keeping the argument content unchanged.	Compare judge scores before and after the order swap.
Country-Label Test	Anonymise or swap country labels while keeping the argument content constant.	Compare judge scores across the original and modified country labels.
Repeated-Judgment Test	Provide the judge with the same cached debate multiple times under identical conditions.	Measure variation in judge scores across repeated evaluations.

5.4 Recommender Evaluation

The recommendation is evaluated in two ways: by *what settlement it lands on*, scored against the benchmark computed in Section 4, and by *how it reads*, scored on a rubric. The first is arithmetic and exact; the second needs a reader. Both are reported, because a recommendation can be well written and still leave value on the table.

Table 8: Outcome metrics, computed by enumeration over the 81 settlements.

Metric	What it measures
Joint value recovered	Joint utility as a fraction of the maximum available. The primary metric for the research question.
Integrative gain	Joint value above the split-the-difference baseline, as a fraction of what was available above it. Zero means the process did no better than halving every issue; one means it found the best trade.
Pareto optimality	Whether any other settlement would have been better for both parties. Being off the frontier means value was left unclaimed.
Nash ratio	Measured against the balanced benchmark, so a lopsided settlement cannot score well on joint value alone.
Singapore's utility	Singapore's utility and its gain over split-the-difference: the number that answers the user's question rather than the marker's.

These are computed by enumeration rather than estimated, so for a given recommended package they are exact and reproducible from the scenario file alone; variation across repeated runs is reported in Section 5.6.

Comparison across conditions. B0, B1 and B2 are compared on the metrics above, paired scenario by scenario since all three receive identical input, with the call counts of Table 3 reported alongside every comparison.

Agreement with the historical settlement. For the historical cases we also compare the recommended package with what the parties actually settled on, issue by issue. Two cautions

govern how this is read. First, the issue set was authored by people who knew how the negotiations ended, so absolute agreement is partly an artefact of construction and is reported as contaminated; the valid question is whether B2 matches history more often than B0, since all conditions face the same menu. Second, the historical settlement is not a target to be hit. Scored against our own frontier the 2015–16 haze outcome is worth 0.268 to Singapore and 0.660 to Indonesia, is not Pareto-optimal, and sits 0.430 below the best jointly available, so a recommendation that *beats* history is a better result than one that reproduces it.

Rubric. Separately, the recommendation is scored 1–5 on feasibility, coverage of major trade-offs, consistency with Singapore's priorities, consideration of counterpart interests, and reflection of strong arguments identified during the debate. The overall rubric score is

$$\text{Overall Score} = \frac{1}{5}\sum_{i=1}^{5} s_i,$$

where s_i is the score for criterion i. We also assess whether the conditional options are genuinely distinct, rather than rewordings of one recommendation, and whether semantically equivalent policies expressed in different wording are being counted as different.

5.5 Comparison with Single-Model Baseline

This comparison addresses the main research question. The baseline (B0) gives a single LLM the same scenario, briefs, interests and constraints as the full system (B2) and asks for a recommendation directly, with no debate, judge or rounds. B1 runs the debate without a judge. All conditions are defined in Table 3.

Measures. All conditions are scored on both measures in Table 9. Neither is reported alone: a recommendation can read well yet leave joint value unclaimed, while outcome metrics alone cannot show whether the advice is usable.

Table 9: Measures used to compare B0, B1 and B2.

Measure	What is scored
Outcome metrics (Section 5.4)	Joint value recovered (primary); integrative gain over the split-the-difference baseline
Recommendation rubric (1–5, Section 5.4)	Feasibility; coverage of major trade-offs; consistency with Singapore's priorities; consideration of counterpart interests; reflection of strong arguments from the debate

Protocol. Comparisons are paired by scenario and repeated across runs (Section 5.6). Since B2 uses eleven model calls against B0's two, a B2 advantage may reflect compute alone. B1, with eight calls and no judge, isolates the contribution of adjudication.

Negative result. If B2 does not recover more joint value than B0, or recovers no more than B1, structured debate has not justified its cost. The precomputed frontier makes this outcome

measurable rather than subjective.

5.6 Repeated Runs and Aggregate Evaluation

Each selected crisis scenario is run multiple times under identical model and generation settings.

Table 10: Aggregate evaluation across repeated experimental runs.

Aggregate evaluation	Calculation / comparison	Reported result
Within-Scenario Performance	For each crisis scenario, average each evaluation metric across repeated runs.	Mean ± SD for each metric per scenario.
Overall Performance	Average each evaluation metric across all crisis scenarios and repeated runs.	Overall mean ± SD for each metric.
Multi-Agent Improvement	For each matched scenario and run, calculate the difference between the multi-agent and single-agent scores: $\Delta = S_{\text{multi}} - S_{\text{single}}$.	Mean score difference (Δ) for each recommendation metric.
Improvement Consistency	Count the number of matched runs in which the multi-agent system achieves a higher score than the single-agent baseline.	Percentage of runs in which the multi-agent system scores higher.
Scenario Sensitivity	Compare mean performance scores across the two crisis scenarios.	Per-scenario mean ± SD and range of scores across scenarios.
Run Stability	Calculate the coefficient of variation (CV) for repeated runs of the same scenario.	$\text{CV} = \frac{\text{SD}}{\text{Mean}}$; lower values indicate greater stability.
Failure Rate	Count runs containing predefined failures, such as major constraint violations, role inconsistency, or unsupported recommendations.	$\frac{\text{Failed runs}}{\text{Total runs}} \times 100\%$.

We also include representative successful and unsuccessful trajectories to explain behaviours that aggregate scores may not capture. This reduces the likelihood that conclusions are based on an unusually good or poor individual LLM run.

6. Risks, Limitations, and Responsible Practice

6.1 Risks

Hallucinated or unsupported information. Agents may introduce inaccurate facts, fabricated evidence, or unsupported assumptions during debate. Incorrect information introduced by one agent may subsequently influence the other agent, the judge, and the recommender. Where external evidence is used, agents will be grounded in an approved set of sources or a retrieval mechanism, and source-backed claims will be traceable where feasible.

Overreliance on the judge agent. The system relies on an LLM judge to evaluate the arguments from the country-agents before its assessments are passed to the recommender. Errors

or biases in the judge's evaluation may directly influence the final recommendation. As such, the judge should not be treated as an objective authority, and its consistency and robustness will be evaluated separately.

Error propagation across agents. Errors can accumulate across the sequential pipeline from the country-agents to the judge to the recommender. We will therefore evaluate intermediate outputs from the country-agents in addition to the final recommendation, allowing failures to be traced to individual components.

6.2 Limitations

Complexity of representing national interests. Even with detailed prompts, retrieval, and multiple rounds of debate, representing the full complexity of a country's interests through a single LLM agent remains inherently limited. National interests are shaped by multiple government institutions, political actors, economic conditions, public preferences, diplomatic relationships, and changing circumstances that cannot be comprehensively represented through a fixed agent profile. Therefore, even a technically robust system cannot be expected to reproduce the depth of contextual knowledge, institutional experience, and judgment available to experienced policymakers and government bodies.

Historical scenarios as imperfect ground truth. Historical crises such as the haze (2015–16) and COVID-19 (2020) cases used here provide documented mitigation measures that can serve as reference points for evaluation. However, historical policy responses are not necessarily the uniquely correct or optimal responses; they should therefore be treated as reference ground truth for comparison rather than absolute ground truth. Policies developed for a particular period, region, political environment, or economic context may also be less relevant to present-day or future crises.

Scope and generalisability. Evaluation will cover only a limited number of crises, countries, and negotiation configurations due to project time and computational constraints. Performance on these scenarios may not generalise to other crisis types, regions, countries, models, or geopolitical contexts.

6.3 Responsible Practice

The project will use public or appropriately licensed information and will not include private, confidential, or personally identifiable information in agent prompts, retrieved evidence, or evaluation materials. Prompts, evaluation criteria, model configurations, and major experimental assumptions will be documented where feasible to improve transparency and reproducibility.

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